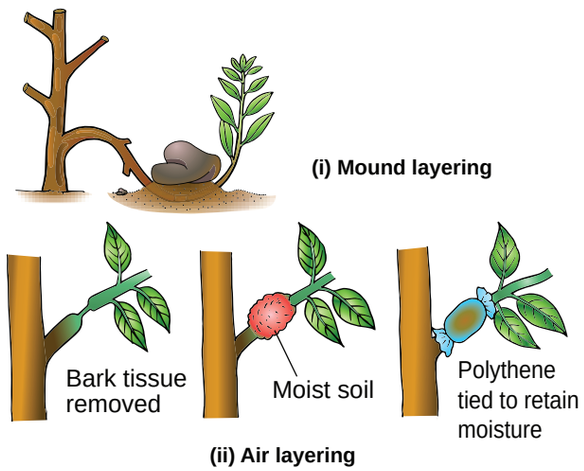


a) Types of Grafting

Figure 1.2(a): Artificial methods of vegetative reproduction in plants

the rooted part is cut and planted to grow as a new plant. Examples: *Ixora* and *Jasminum*. Mound layering and Air layering are few types of layering (Figure 1.2 b).

i. Mound layering: This method is applied for the plants having flexible branches. The lower branch with leaves is bent to the ground and part of the stem is buried in the soil and tip of the branch is exposed above the soil. After the roots emerge from the part of the stem buried in the soil, a cut is made in parent plant so that the buried part grow into a new plant.



b) Types of Layering

Figure 1.2(b): Artificial methods of vegetative reproduction in plants

ii. Air layering: In this method the stem is girdled at nodal region and hormones are applied to this region which promotes rooting. This portion is covered with damp or moist soil using a polythene sheet. Roots emerge in these branches after 2-4 months. Such branches are removed from the parent plant and grown in a separate pot or ground.

Advantages of conventional methods

- The plants produced are genetically uniform.
- Many plants can be produced quickly by this method.
- Some plants produce little or no seeds; in others, the seeds produced do not germinate. In such cases, plants can be produced in a short period by this method.
- Some plants can be propagated more economically by vegetative propagation. Example: *Solanum tuberosum*.
- Two different plants with desirable characters such as disease resistance and high yield can be grafted and grown as a new plant with the same desirable characters.

Disadvantages of conventional methods

- Use of virus infected plants as parents produces viral infected new plants.
- Vegetative structures used for propagation are bulky and so they are difficult to handle and store.

1.3 Sexual Reproduction

In previous classes reproduction in lower plants like algae and bryophytes was discussed in detail. Sexual reproduction involves the production and fusion of male and female gametes. The former is called gametogenesis and the latter

TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_001-032 CH 01.indd 5 3/14/2020 10:30:01 AM